Seminar Condensed Mathematics: Locally Compact Abelian Groups

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Outline

1. Structure of locally compact abelian groups
2. The key computation for RHom
3. Derived categories

Structure of locally compact abelian groups

The statements

If A is a locally compact abelian group then

- $A \simeq \mathbb{R}^n \times A'$ where A' admits a compact open subgroup.
- *Pontrjagin duality*: $A \to \mathbb{D}(\mathbb{D}(A))$ is an isomorphism, where $\mathbb{D}(A) := \mathrm{Hom}(A, \mathbb{T})$ and $\mathbb{T} := \mathbb{R}/\mathbb{Z}$.
- $\mathbb{D}$ takes compact abelian groups to discrete abelian groups and vice versa.

Hom

Proposition

There exists a natural isomorphism of condensed abelian groups

$$\underline{\mathrm{Hom}}(\underline{A}, \underline{B}) \simeq \underline{\mathrm{Hom}(A, B)},$$

where we assume further that A, B are Hausdorff and A is compactly generated.

Proof

Step 1: Construct a map

$$\underline{\mathrm{Hom}}(\underline{A}, \underline{B}) \to \underline{\mathrm{Hom}(A, B)}.$$

Unwinding definition,

$$\mathrm{Hom}(\underline{A}, \underline{B})(S) = \mathrm{Hom}(\underline{A} \otimes \mathbb{Z}[S], \underline{B}).$$

Each element in $\mathrm{Hom}(\underline{A} \otimes \mathbb{Z}[S], \underline{B})$ corresponds to a map $S \to \mathrm{Hom}(A, B)$. This map is continuous since the surjection

$$\mathbb{Z}[\underline{A} \times S] \to \underline{A} \otimes \mathbb{Z}[S]$$

induces a map of condensed sets $\underline{A \times S} = \underline{A} \times S \to \underline{B}$.

Hom

Step 2:

- Injectivity: Obvious.
- Surjectivity: Consider the exact sequence

$$\mathbb{Z}[\underline{A} \times \underline{A}] \xrightarrow{[(a_1,a_2)]\mapsto[a_1+a_2]-[a_1]-[a_2]} \mathbb{Z}[\underline{A}] \to \underline{A} \to 0.$$

We obtain the exact sequence

$$\mathbb{Z}[\underline{A} \times \underline{A}] \otimes \mathbb{Z}[S] \to \mathbb{Z}[\underline{A}] \otimes \mathbb{Z}[S] \to \underline{A} \otimes \mathbb{Z}[S] \to 0$$

It suffices to show that

$$\mathbb{Z}[\underline{A} \times \underline{A}] \otimes \mathbb{Z}[S] \to \mathbb{Z}[\underline{A}] \otimes \mathbb{Z}[S] \to \underline{B}$$

is 0. This is clear since it is given by
$(a_1, a_2, s) \mapsto s(a_1 + a_2) - s(a_1) - s(a_2)$.

The key computation for RHom

The key computation

Let $A = \prod_I \mathbb{T}$. Then

- $\underline{R\mathrm{Hom}}(A, M) = \bigoplus_I M[-1]$ for any discrete abelian group M.
- $\underline{R\mathrm{Hom}}(A, \mathbb{R}) = 0$.

Remark

Using the exact sequence

$$0 \to \mathbb{Z} \to \mathbb{R} \to \mathbb{T} \to 0,$$

we obtain the distinguished triangle

$$\underline{R\mathrm{Hom}}(\mathbb{T}, \mathbb{R}) \to \underline{R\mathrm{Hom}}(\mathbb{R}, \mathbb{R}) \to \underline{R\mathrm{Hom}}(\mathbb{Z}, \mathbb{R}) \to \underline{R\mathrm{Hom}}(\mathbb{T}, \mathbb{R})[1]$$

from which we deduce that $\underline{R\mathrm{Hom}}(\mathbb{R}, \mathbb{R}) = \mathbb{R}$.

The key computation for RHom

The theorem of Eilenberg-MacLane, Breen, Deligne

There exists functorial resolution

$$\to \bigoplus_{j=1}^{n_i} \mathbb{Z}[A^{r_{i,j}}] \to \cdots \to \mathbb{Z}[A^3] \oplus \mathbb{Z}[A^2] \to \mathbb{Z}[A^2] \to \mathbb{Z}[A] \to A \to 0$$

for an abelian group A.

Corollary

There exists a functorial spectral sequence

$$E_1^{i_1,i_2} = \prod_{j=1}^{n_{i_1}} H^{i_2}(A^{r_{i_1,j}} \times S, M) \Rightarrow \underline{\mathrm{Ext}}^{i_1+i_2}(A, M)(S)$$

in abelian groups A, M and extremally disconnected set S.

The key computation for RHom

Proof of the corollary

Given a ringed site $(\mathcal{C}, \mathcal{O})$ and any bounded above complex of $\mathcal{O}$-modules $A^\bullet$, there exists a spectral sequence

$$E_1^{i,j} = \mathrm{Ext}^j(A^{-i}, M) \Rightarrow \mathrm{Ext}^{i+j}(A^\bullet, M).$$

In particular, for the ringed site $(\mathcal{C}_{/U}, \mathcal{O})$, there exists a spectral sequence

$$E_1^{i,j} = \mathrm{Ext}^j(A^{-i}|_U, M|_U) \Rightarrow \mathrm{Ext}^{i+j}(A^\bullet|_U, M|_U).$$

In general, the ext-sheaf $\underline{\mathrm{Ext}}^j(A, M)$ is the sheafification of the construction $U \mapsto \mathrm{Ext}^j(A|_U, M|_U)$

The key computation for RHom

In our case $U = S$ is an extremally disconnected set so

$$E_1^{i_1,i_2} = \underline{\mathrm{Ext}}^{i_2}(\bigoplus_{j=1}^{n_i} \mathbb{Z}[A^{r_{i_1,j}}], M)(S) \Rightarrow \underline{\mathrm{Ext}}^{i_1+i_2}(A, M)(S).$$

The result follows as

$$\underline{\mathrm{Ext}}^{i_2}(\bigoplus_{j=1}^{n_i} \mathbb{Z}[A^{r_{i_1,j}}], M)(S) \simeq \prod_{j=1}^{n_i} \underline{\mathrm{Ext}}^{i_2}(\mathbb{Z}[A^{r_{i_1,j}}], M)(S)$$

$$\simeq \prod_{j=1}^{n_i} \underline{\mathrm{Ext}}^{i_2}(\mathbb{Z}[A^{r_{i_1,j}}] \otimes \mathbb{Z}[S], M) \simeq \prod_{j=1}^{n_i} \underline{\mathrm{Ext}}^{i_2}(\mathbb{Z}[A^{r_{i_1,j}} \times S], M)$$

$$= \prod_{j=1}^{n_i} H^{i_2}(A^{r_{i_1,j}} \times S, M).$$

The key computation for RHom

Proof of the key computation

Proof of (i):

- *Step 1*: Assume I is finite and moreover I is single. One needs to show that

$$M[-1] = \underline{R\mathrm{Hom}}(\mathbb{Z}[1], M) \to \underline{R\mathrm{Hom}}(\mathbb{T}, M)$$

is an isomorphism. Using the distinguished triangle

$$\underline{R\mathrm{Hom}}(\mathbb{R}, M)[-1] \to \underline{R\mathrm{Hom}}(\mathbb{Z}[1], M) \to \underline{R\mathrm{Hom}}(\mathbb{T}, M)$$

$$\to \underline{R\mathrm{Hom}}(\mathbb{R}, M),$$

it suffices to show that $\underline{R\mathrm{Hom}}(\mathbb{R}, M) = 0$.

The key computation for RHom

Using the spectral sequence in the corollary, it reduces to show that

$$H^i(\mathbb{R}^r \times S, M) \to H^i(S, M)$$

is an isomorphism for any $r \geq 0$. As the construction $S \mapsto \mathbb{Z}[S]$ preserves colimits and using Stacks project Tag0949 (for the filterd colimits in $\mathrm{Cond}(\mathrm{Ab})$ are exact), we obtain

$$H^i R\Gamma(\mathbb{R}^r \times S, M) = H^i R \varprojlim_n R\Gamma([-n, n]^r \times S, M).$$

Hence, it suffices to show that each

$$H^i([-n, n]^r \times S, M) \to H^i(S, M)$$

is an isomorphism. This follows from Theorem 3.2.

The key computation for RHom

- *Step 2*: The result follows for any infintie set I since

$$\varinjlim_{\text{finite } J\subset I} \mathrm{R}\underline{\mathrm{Hom}}(\prod_J \mathbb{T}, M) \simeq \mathrm{R}\underline{\mathrm{Hom}}(\prod_I \mathbb{T}, M).$$

Proof of (ii):

- *Step 1*: We again use

$$\mathrm{R}\underline{\mathrm{Hom}}(A, \mathbb{R})(S) = \mathrm{RHom}(\mathbb{Z}[A \times S], M)$$

and note that each term in the resolution of $\mathbb{Z}[A \times S]$ is acyclic and represented by $C(-, \mathbb{R})$ (Theorem 3.3). Hence, $\mathrm{R}\underline{\mathrm{Hom}}(A, \mathbb{R})(S)$ is represented by the complex

$$0 \to \oplus_{j=1}^{n_0} C(A^{r_{0,j}} \times S, \mathbb{R}) \to \oplus_{j=1}^{n_1} C(A^{r_{1,j}} \times S, \mathbb{R}) \to \cdots$$

The key computation for RHom

- *Step 2*: By Proposition 4.17, for any $n \geq 1$, the multiplication by n of the above complex and the morphism of complexes $[n]$ induced by multiplication by n on A are homotopic via a homotopy $h_\bullet$. Set $n = 2$.
Let f is in the degree i component with $df = 0$. We want to show that $f \in \mathrm{Im}(d)$. We have $f = \frac{1}{2}[2]f + d(\frac{1}{2}h(f))$. Hence

$$f = \frac{1}{2^n}[2^n](f) + d\left(\frac{1}{2}h(f) + \frac{1}{4}h([2]f) + \cdots + \frac{1}{2^n}h([2^{n-1}]f\right).$$

The result follows by taking the limit as $n \to +\infty$.

Derived categories

Some terminologies

- LCA is the category of locally compact abelian groups.
- A morphism $f : A \to B$ is *strict* if $A/\ker(f) \to \mathrm{Im}(f)$ is an isomorphism.
- A complex is called *strictly exact* if it is exact and its differentials are strict morphisms.
- $D^b(\mathrm{LCA}) := \mathrm{Ch}^n(\mathrm{LCA})[W^{-1}]$ where W^{-1} consists of morphisms of complexes whose cones are strictly exact.

Derived categories

Corollary

There exists a natural fully faithful functor

$$D^b(\mathrm{LCA}) \to D(\mathrm{Cond}(\mathrm{Ab})).$$

Proof

Step 1: One needs to show that a strict exact sequence

$$0 \to A \to B \to C \to 0$$

induces an exact sequence

$$0 \to \underline{A} \to \underline{B} \to \underline{C} \to 0.$$

Since $\mathrm{LCA} \to \mathrm{Cond}(\mathrm{Ab})$ commutes with limits, it suffices to show that $\underline{B} \to \underline{C}$ is surjective.

Derived categories

Note that $B \to C$ is a open map. Let $h \in \underline{C}(S)$ for an extremally disconnected set S. Since S is compact, $h(S)$ is compact. Let K be a compact neighborhood of 1_B. Since $h(S)$ is compact, we obtain a finite cover

$$h(S) \subseteq \bigcup_i (c_i + g(K)).$$

Let $b_i \in B$ such that $c_i = g(b_i)$ and let $Z = \bigcup_i b_i + K$. We have that $h(S) \subseteq g(Z)$. Since S is a projective object in CHaus, there exists a morphism $h' : S \to Z$ such that $g|_Z \circ h' = h$. It follows that $\underline{B}(S) \to \underline{C}(S)$ is surjective.

Derived categories

Step 2: We show that

$$\mathrm{RHom}_{\mathrm{LCA}}(A, B) \to \mathrm{RHom}_{\mathrm{Cond(Ab)}}(\underline{A}, \underline{B})$$

is an isomorphism for any $A, B \in \mathrm{LCA}$. By the result the structure of locally compact abelian groups, the statement is reduces to the case $A = \mathbb{R}$, discrete, or compact abelian groups. The case discrete reduces to the case $A = \mathbb{Z}$, which is trivial. The case $\mathbb{R}$ reduces to the case $A = \mathbb{T}$. We assume that A is compact. Using Pontrjagin duality, we can assume that $A = \prod_I \mathbb{T}$.

Derived categories

Similarly, we can assume $B = \prod_J \mathbb{T}$ and moreover $B = \mathbb{T}$. This case will follow from the case $B = \mathbb{Z}$ or $B = \mathbb{R}$. The results follows as we computed $\mathrm{RHom}_{\mathrm{Cond(Ab)}}(\underline{A}, \underline{B})$, and $\mathrm{RHom}_{\mathrm{LCA}}(A, B)$ is computed by N. Hoffmann and M. Spitzweck and coincides with our computation.